



EAST WEST UNIVERSITY

LAB REPORT

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Course Name: Electronic Circuits

Course Code: CSE 251

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Group No: 10A

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Experiment Title

To Study the I-V Characteristics of a Zener Diode

Objective :

- To study the relationship between the current and voltage in a Zener diode under forward and reverse bias conditions.
- To identify the Zener breakdown voltage at which the diode conducts in reverse bias.
- Have a working understanding of circuit variables and circuit components.
- Be familiar with Zener diode and how to determine p-n junction current.
- To confirm I-V Characteristic of Zener diode

Theory:

A Zener diode is a special type of diode. In forward bias, it acts like a regular diode and allows current to flow once the voltage reaches about 0.7V. However, in reverse bias, it is different. When the reverse voltage goes beyond a certain value, called the Zener breakdown voltage, the diode starts conducting in reverse without getting damaged. This property makes it useful for keeping the voltage steady in circuits.

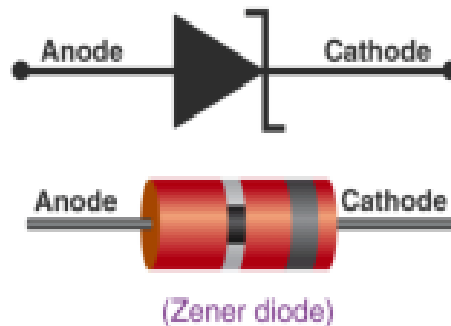


Figure 1: Zener Diode

- **Forward Bias:** Current flows through the diode once the forward voltage reaches around 0.7V (for silicon).
- **Reverse Bias:** When reverse voltage exceeds a specific value (the Zener breakdown voltage), the diode conducts without damage. This unique property makes it ideal for voltage regulation.

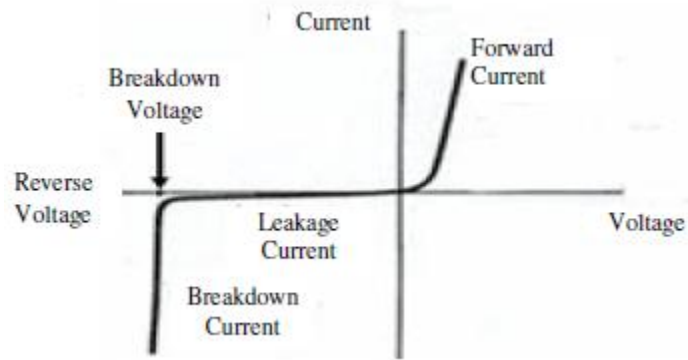


Figure 2: Zener breakdown and Avalanche of diode

Apparatus Required:

1. Zener Diode (1N4007)
2. Resistor (2.2K)
3. Breadboard
4. Connecting wires
5. Multimeter
6. DC power supply
7. Ammeter

Working Diagrams:

Forward Bias:

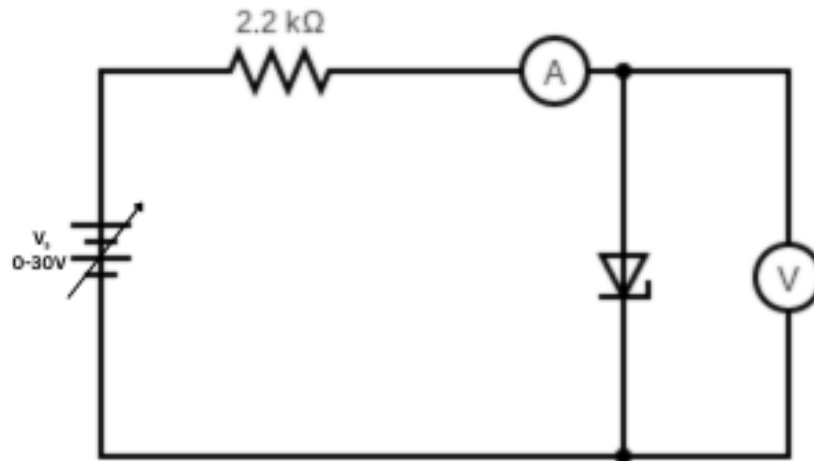


Figure 3: Zener Diode in Forward Bias

Reverse Bias:

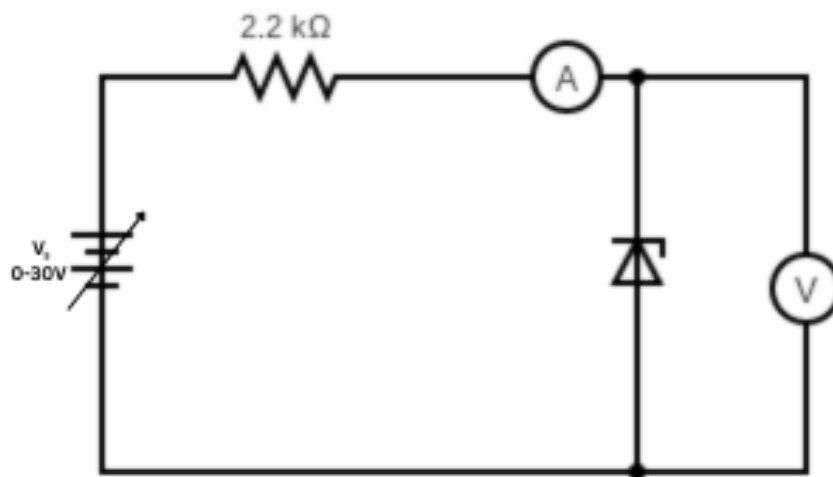


Figure 4: Zener Diode in Reverse Bias

Data Table:**Forward Bias:**

No. of observation	V_z (volt)	I_z (mA)	V_s
1	0	0	0
2	0.5	0	0.5
3	0.55	0	0.6
4	0.6	0	0.67
5	0.70	0.2	1.5
6	0.75	1.4	4.4
7	0.76	2.0	5.6
8	0.77	2.6	7.1
9	0.78	3.6	9.9
10	0.79	5.0	12.3
11	0.80	7.3	17.4
12	0.805	9.0	21.1
13	0.808	10	23.2

Reverse Bias:

No. of observation	-V_z (volt)	-I_z (mA)	V_s
1	0	0	0
2	0.15	0	0.2
3	0.35	0	0.4
4	0.53	0	0.6
5	0.74	0	0.8
6	0.9	0	1
7	1.08	0	1.2
8	1.34	0	1.3
9	1.57	0	1.6
10	3.4	0	3.6
11	4	0	4.6
12	4.2	0.2	5.6
13	5.94	0.4	7.2
14	5.96	1.4	9.3
15	5.98	4.2	15.4
16	6	6.0	19.9
17	6.016	7.4	22.3
18	6.023	8.4	24.5
19	6.032	9.1	26.05
20	6.035	9.2	26.28
21	6.037	9.7	27.38
22	6.040	10	28.04

Graph

Forward Bias:

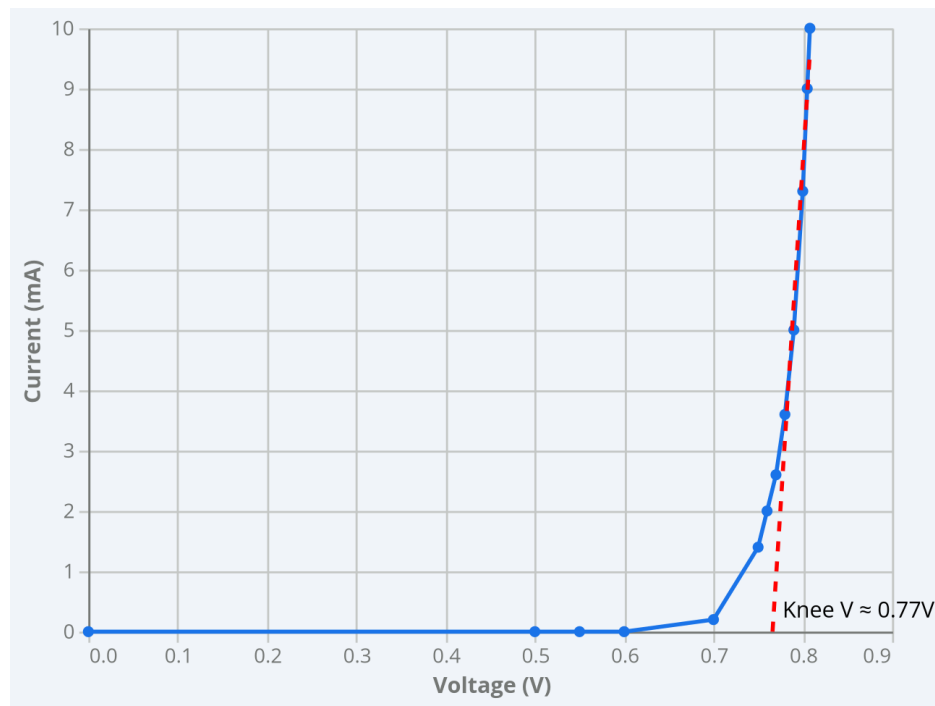


Figure 5: Zener Diode in Forward Bias

Reverse Bias:

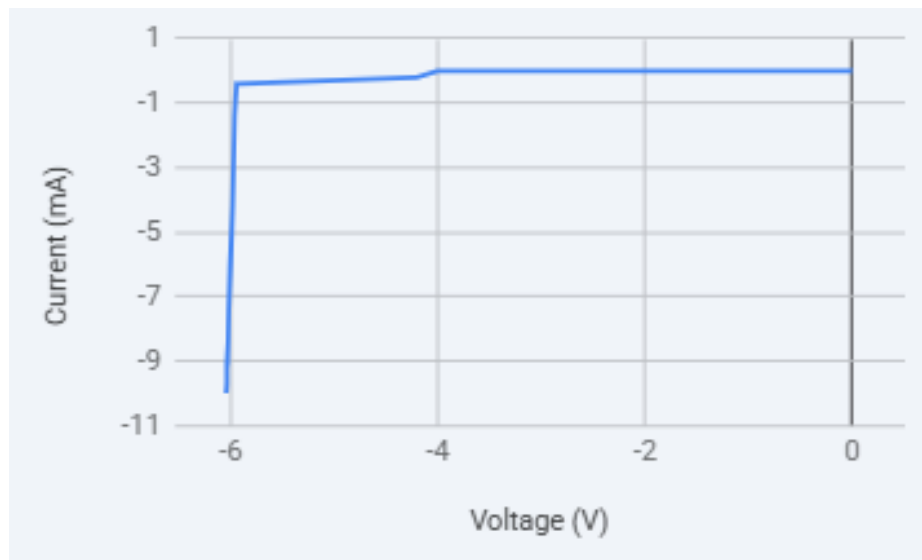


Figure 6: Zener Diode in Reverse Bias

Results

The I–V characteristics of the diode were successfully observed under both forward and reverse bias conditions. In the forward bias region, the diode showed negligible current at low applied voltage. The current started to increase significantly after approximately $V_D \approx 0.75$. This point represents the knee (threshold) voltage of the diode. The experimentally determined knee voltage is approximately 0.77, which is close to the theoretical value of 0.7 V for a silicon diode.

In the reverse bias condition, the diode initially blocked current flow, showing nearly zero current. However, at a reverse voltage of approximately -5.9 V, the diode entered the breakdown region, and the current increased sharply. This value represents the breakdown voltage of the diode.

Discussion:

The experimental results show good agreement with the theoretical behavior of a diode. In forward bias, the current remains very small until the applied voltage reaches the threshold level. After this point, a small increase in voltage produces a large increase in current, confirming the exponential I–V relationship of the diode.

The slight difference between the theoretical knee voltage (0.7 V) and the measured value (0.77 V) may be due to instrument limitations, internal resistance of the diode, contact resistance in the circuit, and human reading errors during measurements.

In reverse bias, the diode effectively blocked current until the breakdown voltage was reached. After breakdown, the current increased rapidly, indicating that the diode no longer behaves as an insulator in this region. The observed breakdown voltage of approximately -5.9 V may vary slightly from the theoretical value due to manufacturing tolerances and experimental conditions.

Overall, the experiment successfully demonstrated the rectifying behavior of the diode and verified its I–V characteristics in both forward and reverse bias conditions.